
The Wall Is Disjunction, Not Abstraction: Sparse Autoencoders on a Game-Playing Network

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Sparse autoencoders (SAEs) trained on game-playing networks reliably recover
2 per-cell board facts and reliably fail on relational concepts such as “this line is one
3 move from completion.” Whether the network lacks the concept or the dictionary
4 cannot name it is undecidable from one layer with nothing to compare against. We
5 measure a linear probe and an SAE dictionary on the same activations at two layers
6 of a Quarto agent, plus a third measurement of where a concept goes when no single
7 atom carries it. Across three agents, three SAE architectures, four dictionary widths
8 and 546 ground-truth concepts: (i) the preferred layer partitions by concept family,
9 so a pooled comparison averages effects of opposite sign and reports nothing;
10 (ii) the probe overturns the natural reading of that split, since for state threats the
11 convolutional layer is equally or more linearly available in 11 of 12 cells while
12 its dictionary isolates $1.4\text{--}4.8\times$ less; and (iii) the residual failure has a specific
13 logical shape. Holding basis, agent-relativity and base rate fixed and varying only
14 logical form, single-atom capture drops from 81.2% to 0.0% the moment a concept
15 is a disjunction; agent-relativity, the property we had previously blamed, costs 2
16 points. The dictionary instead rebuilds each 4-way disjunction from exactly its four
17 disjuncts (modal cost 4 latents, and none of 266 concepts is cheaper), and an $8\times$
18 larger dictionary does not move it. The wall is not missing information and not
19 capacity; it is the absence of an atom for OR.

20 1 Introduction

21 Sparse autoencoders have become the default tool for decomposing neural activations into human-
22 readable units [Cunningham et al., 2023, Gao et al., 2024]. Board-game agents are an attractive
23 testbed because the ground truth is exhaustive and cheap: any property of a position can be computed
24 exactly, so a dictionary can be scored against a full concept menu rather than against post-hoc human
25 labels [Karvonen et al., 2024, Li et al., 2022, Nanda et al., 2023].

26 Work in this setting keeps reporting the same asymmetry. Dictionaries recover *per-cell* facts (which
27 square holds a piece, which attributes it has) at high fidelity, and recover *relational* facts (which line is
28 one move from being completed) poorly. The asymmetry is robust across architectures and capacities,
29 which makes it look structural rather than incidental. Two very different explanations predict exactly
30 this observation. On **the information account**, the network never forms the relational concept; it
31 plays well by some non-decomposable shortcut, so no dictionary can surface what is not there, and
32 the fix is to change the *model*: different objective, architecture, or auxiliary supervision. On **the**
33 **factorisation account**, the network does form the concept but distributes it over many directions,
34 so no single atom aligns with it [Engels et al., 2024, Park et al., 2024], and the fix is to change the
35 *dictionary*: units that are not one-dimensional.

These prescriptions are opposite, and the standard evaluation cannot tell them apart, because it measures one dictionary at one layer and has no reference for what was available to be decomposed. Our first step is a measurement that is cheap and, to our knowledge, not standard: run a linear probe *and* an SAE on the *same* activations, at two layers, for the same concepts and the same positions. The probe upper-bounds what is linearly present (*availability*); the best single latent measures what the dictionary presents as one unit (*isolation*).

That comparison settles which account is right but not what to do about it, so we add a second measurement. When no single atom carries a concept the signal is still somewhere: we regress the concept on the 64 most associated latents and ask how much they recover at all (*retention*, against a matched dictionary trained on an untrained network) and what share of it sits in the single best one (*concentration*). Together these turn “the SAE score is low” into “the dictionary lost it” versus “the dictionary has it, in pieces”, and since we know what each concept is a function of, we can then ask which pieces. Our contributions:

1. **The preferred layer partitions by concept family** (§4.1): the cell-averaged gap points the same way for all fifteen families over nine matched (agent $\times$ recipe) cells: per-cell toward the convolutional layer, relational toward the bottleneck; the fifteenth, the trivial-baseline family, is a near-tie that splits by agent. A pooled comparison averages a $+0.28$ against a -0.12 and is uninterpretable.
2. **The gap is factorisation, not information** (§4.2): for state threats the convolutional layer is equally or *more* linearly available in 11 of 12 cells, yet its dictionary isolates $1.4\text{--}4.8\times$ less.
3. **Retention separates two failures that score alike** (§4.3): state threats at the strongest agent’s convolutional layer and disjunctive concepts at its bottleneck both fall far short of a single atom (concentration 0.13 and 0.31 against the 0.70 threshold), but retention reads the first as barely in the dictionary ($2.3\times$ the untrained-network floor) and the second as fully in it (median $R^2 = 0.741$) and split across its disjuncts.
4. **The residual wall is a disjunction wall** (§4.4): within a single basis, with agent-relativity and base rate held fixed, capture goes $81.2\% \rightarrow 0.0\%$ when the only change is that the concept ORs over four attribute poles. Agent-relativity, which our own earlier reading blamed, costs 2 points.
5. **The mechanism, read directly, plus a capacity control** (§4.5): the dictionary rebuilds a 4-way disjunction from exactly four latents (the mode is 4 and *none* of the 266 concepts is below 4), and growing the dictionary $8\times$ ($4,096 \rightarrow 32,768$) leaves the same 22 of 23 concepts spread.

2 Setup

Game. Quarto is played on a 4×4 board with 16 distinct pieces carrying four binary attributes each (tall/short, black/white, square/round, holed/solid). A player wins by completing a line (or, in the variant we use, a 2×2 square) of four pieces sharing at least one attribute. The distinctive rule is that *your opponent chooses the piece you must place*, which makes “is the piece I am about to hand over a winning piece for my opponent” a first-class strategic concept and cleanly separates board-state from agent-relative concepts.

Agents. Three value-based convolutional agents share the same trunk and differ mainly in training procedure: **M1** selects self-play moves with a depth-2 minimax selector, **M2** distils a depth-2 minimax oracle for selection and keeps it enabled throughout training, and **M3** adds dense piece-safety shaping. **M3** also carries one training-only auxiliary head that is never read at inference, and **M1** trains for 4,350 epochs against 10,000 for **M2/M3**. Head-to-head, $M3 > M2 > M1$, so the three form an ordered competence axis; all three remain below a depth-2 minimax baseline. Each contributes its own deduplicated self-play position set (290,147 / 289,795 / 296,045 positions), aggregated over four opponent-mode mixtures and deduplicated *after* aggregation.

Layers. We hook conv2, the second convolutional block, flattened from (32, 4, 4) to 512 dimensions, and fc1, the 512-unit bottleneck that follows it. The two are **the same width**, so SAEs at both hooks take identical input dimension and dictionary size: the comparison is not confounded

by capacity, which is unusual for a cross-layer study and is what makes the matched design below possible.

Concepts. Ground truth is a menu of 546 board-state properties in four *bases*: STATE, a state-only menu (164 concepts); COUNT, a threat-count reframing of the same facts (173); COUNT[−], those same 173 concepts stated on the opposite (negative) attribute poles; and AGENT-REL., phrased from the mover’s point of view (36). Bases are alternative *packagings*; the comparable unit across them is the *concept family* (board_occupancy, board_attribute, line_threat, square_threat, ...), stamped on every schema entry. We group families into **per-cell** (a fact about one square) and **relational** (a fact about a set of squares, or about the mover). Whole-basis averages are not interpretable (the bases partition the menu differently), so every comparison below is per family. COUNT[−] earns its place in §4.5: a Quarto line wins on *any* shared value, so “short” wins exactly as “tall” does, and each agent-relative threat concept is exactly the OR of its COUNT and COUNT[−] counterparts (verified with zero violations), which is what fixes its disjunct count at eight rather than four.

SAEs. The probe comparison (§4.1–4.2) uses three recipes trained at *both* hooks: TopK ($k=32$), TopK ($k=64$), BatchTopK ($k=32$) [Gao et al., 2024, Bussmann et al., 2024], all at expansion 8 ($d_{\text{dict}} = 4096$), one seed: three agents $\times$ three recipes = nine **matched cells**, in each of which the hook is the only thing that varies. The retention analysis (§4.3 onward) runs on a separate, larger panel of 114 (dictionary $\times$ basis) cells over the same agents and hooks: the top three *conditions* per cell by threat-family MCC, a 4-condition $\times$ 3-seed replicate grid, and four supervised anchored dictionaries as a positive control; for 12,151 concept verdicts, and additionally covers JumpReLU [Rajamanoharan et al., 2024] and expansions 8/16/32/64.

Metrics. We report Matthews correlation (MCC) as the headline [Chicco and Jurman, 2020]: threat base rates are near 0.01, where F1 moves with the base rate through precision and is not comparable across concepts. We report Youden’s J [Youden, 1950] alongside, since J is prevalence-invariant and MCC is not, and standardise MCC to $p_{\text{ref}} = 0.025$ whenever a comparison crosses populations. Within a matched cell both arms share positions and concept, so prevalence is identical by construction and raw MCC is correct there.

3 Method: availability, isolation, retention, concentration

For a concept c and a layer ℓ : **availability** $A(c, \ell)$ is the MCC of a linear probe on the raw activations of ℓ : what the layer linearly carries before any dictionary, a property of the *model*. **Isolation** $I(c, \ell)$ is the MCC of the single best SAE latent, matched to c over the whole dictionary: what the decomposition presents as *one atom*. **Efficiency** $E = I/A$ is the fraction of available structure the dictionary atomises: $E \approx 1$ means it found what was there, and $E \ll 1$ with high A is the signature of a factorisation failure.

Efficiency says a dictionary failed but not how, so we decompose I . Rank all live latents by their association with c , keep the top $K = 64$, and fit a held-out linear regression of the concept indicator on their codes. **Retention** $R(c, \ell)$ is that fit’s held-out R^2 , reported against R_{rand}^2 , the same quantity for a dictionary trained on the *untrained* network’s activations at the same layer; R is the total signal the dictionary carries however it is spread, and R/R_{rand} is the learned-signal multiple. **Concentration** $S(c, \ell)$ is the share of R the single best latent recovers alone; being a ratio, it separates “spread thinly” from “not there.” Finally k_{90} is the smallest k whose top- k latents reach 90% of R ; it is thresholded nowhere, and §4.5 compares only its *distribution* against a disjunct count known by construction. Each concept is then labelled **captured** ($S \geq 0.70$, over both floors), **absent** (no more signal than the permutation null *or* than the untrained-network floor), or **spread** (above both floors, no single atom carrying it); *captured* is the arm that must fire for the label to mean anything (§4.6).

The decisive quantity is never one number but a pair: I small with A small is an information limit, I small with A large a factorisation limit, and within the latter, S small with R small means the dictionary never picked the concept up while S small with R large means it split it. Two caveats hold throughout: everything here is **decodability, not causality** (matching is greedy argmax, so the matched latent may be a spectator), and A bounds *linear* availability while R is measured over 64 of the 4,096 latents, so both lower-bound presence.

Table 1: Best-latent MCC by concept family, averaged over nine matched (agent $\times$ recipe) cells; $\Delta = \text{fc1} - \text{conv2}$. “fc1 wins” counts cells with $\Delta > 0$. The partition is total: all fifteen cell-averaged gaps point the predicted way, the trivial-baseline family only marginally (-0.008). ‡ base rate far from p_{ref} ; read on raw MCC.

basis	concept family	n	base rate	fc1	conv2	Δ	fc1 wins
state	board_occupancy	16	0.3401	0.448	0.542	-0.094	2/9
state	board_attribute	64	0.1713	0.399	0.518	-0.119	2/9
state	offered_piece_attr	4	0.5042	0.370	0.378	-0.008	3/9
state	line_threat	40	0.0095	0.260	0.102	+0.158	9/9
state	square_threat	36	0.0091	0.388	0.106	+0.281	9/9
state	global_threat‡	1	0.2559	0.383	0.201	+0.183	9/9
state	game_phase	3	0.3333	0.432	0.230	+0.202	9/9
count	line_threat	90	0.0136	0.244	0.110	+0.134	9/9
count	square_threat	81	0.0126	0.340	0.112	+0.228	9/9
count	global_threat‡	2	0.2525	0.360	0.201	+0.159	9/9
agent-rel.	line_threat	10	0.0248	0.195	0.101	+0.094	9/9
agent-rel.	square_threat	9	0.0229	0.273	0.101	+0.172	9/9
agent-rel.	global_threat‡	5	0.4510	0.405	0.217	+0.188	9/9
agent-rel.	pool_reasoning‡	8	0.4928	0.405	0.216	+0.188	9/9
agent-rel.	offered_completion	4	0.0476	0.239	0.147	+0.092	8/9

Table 2: Availability (linear probe) vs. isolation (best latent) on state threats, standardised to $p_{\text{ref}} = 0.025$. conv2 is equally or more linearly available in 11 of 12 cells, while fc1 isolates 1.4–4.8 $\times$ more.

basis / family	agent	availability (LP)			isolation (SAE)		efficiency	
		fc1	conv2	higher	fc1	conv2	fc1	conv2
state/line_threat	M1	0.639	0.815	conv2	0.261	0.185	0.41	0.23
state/line_threat	M2	0.704	0.907	conv2	0.360	0.174	0.51	0.19
state/line_threat	M3	0.893	0.958	conv2	0.550	0.180	0.62	0.19
state/square_threat	M1	0.774	0.886	conv2	0.369	0.192	0.48	0.22
state/square_threat	M2	0.838	0.929	conv2	0.495	0.170	0.59	0.18
state/square_threat	M3	0.938	0.946	conv2	0.851	0.178	0.91	0.19
count/line_threat	M1	0.712	0.840	conv2	0.299	0.189	0.42	0.23
count/line_threat	M2	0.753	0.875	conv2	0.367	0.180	0.49	0.21
count/line_threat	M3	0.873	0.897	conv2	0.483	0.179	0.55	0.20
count/square_threat	M1	0.824	0.873	conv2	0.424	0.196	0.52	0.22
count/square_threat	M2	0.871	0.884	conv2	0.511	0.179	0.59	0.20
count/square_threat	M3	0.890	0.886	fc1	0.702	0.178	0.79	0.20

139 4 Results

140 4.1 The preferred layer partitions by concept family

141 Table 1 gives every family, averaged over the nine matched cells. Per-cell families go to conv2 (fc1
142 wins 2/9, 2/9 and 3/9); every relational family goes to fc1 (9/9 for eleven of twelve, 8/9 for the
143 twelfth). The two groups do not overlap, and the sign is stable across three agents and three recipes,
144 the one exception being the trivial-baseline family, which reverses on M3 and still does so under the
145 repaired conv2 dictionary of §4.6. The methodological consequence is that a *pooled* layer comparison
146 is meaningless here: averaging +0.281 on square threats with -0.119 on cell attributes yields a
147 number whose sign depends on how many concepts of each kind the menu happens to contain, not on
148 anything about the model.

149 4.2 The probe changes what the split means

150 The natural reading of Table 1 is that the bottleneck is where threat information lives. The linear
151 probe says otherwise (Table 2, Figure 1). Over the state-threat cells, **conv2 has the higher linear**
152 **availability in 11 of 12**, while fc1 has the higher efficiency in all 12. The extreme case is agent M3’s

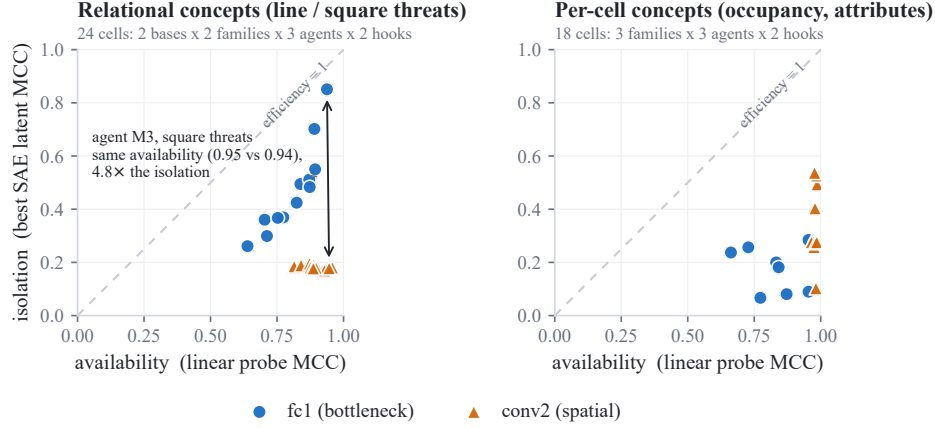


Figure 1: Availability vs. isolation; the dashed diagonal is efficiency = 1. **Left:** relational concepts. Both layers are far to the right (the information is there), but conv2 (triangles) is pinned near $I \approx 0.19$ regardless, while fc1 (circles) climbs toward the diagonal. **Right:** per-cell concepts, where the ordering reverses. The two lowest conv2 points are `offered_piece_attr`, whose base rate of 0.50 makes it the trivial-baseline family.

153 square threats: the two layers carry the concept *equally well* ($A = 0.946$ at conv2 vs. 0.938 at fc1, a
 154 difference of 0.008) and their dictionaries differ by **4.8**× ($I = 0.178$ vs. 0.851).

155 This is the first observation. The information is not lost crossing the bottleneck: a probe recovers it at
 156 0.87–0.96 on *both* sides. What changes is its *form*: distributed across a spatial map at conv2, factored
 157 into individual directions at fc1. Note also how flat the conv2 efficiency column is: 0.18–0.23 across
 158 the twelve cells of Table 2, so whatever prevents atomisation there is insensitive to how much of the
 159 concept is present. Along the competence axis $M1 \rightarrow M2 \rightarrow M3$, by contrast, fc1 efficiency rises
 160 0.48 $\rightarrow$ 0.59 $\rightarrow$ 0.91 on square threats and 0.41 $\rightarrow$ 0.51 $\rightarrow$ 0.62 on line threats, against conv2’s flat
 161 0.22 $\rightarrow$ 0.18 $\rightarrow$ 0.19 and 0.23 $\rightarrow$ 0.19 $\rightarrow$ 0.19. Stronger agents do not merely encode threats *more*;
 162 they encode them in a form a flat dictionary can *name*, and only at the bottleneck.

163 4.3 Retention: two failures that score the same

164 Efficiency treats every low score alike. Retention does not. On the 114-cell panel, pooled over three
 165 bases and all threat concepts:

agent	layer	n	R^2	R^2_{rand}	retention	solo
M1	conv2	810	0.096	0.032	3.0×	0.28
M1	fc1	810	0.163	0.031	5.3×	0.43
M2	conv2	810	0.097	0.032	3.1×	0.25
M2	fc1	810	0.260	0.028	9.2×	0.58
M3	conv2	810	0.087	0.037	2.3×	0.13
M3	fc1	810	0.565	0.037	15.2×	0.92

167 At conv2 the dictionary recovers 2.3–3.1× the untrained-network floor and concentrates 0.13–0.28
 168 of it, for *every* agent; at fc1 both terms climb with competence, to 15.2× and 0.92 on M3. Read
 169 against §4.2, where a probe reads state threats off raw conv2 activations at MCC 0.87–0.96, this says
 170 something sharper than “conv2’s dictionary is worse”: **the concept is in the layer and is not in the**
 171 **dictionary**, not even spread over 64 latents. The convolutional failure is an encoding failure; only
 172 the bottleneck failure is a factorisation failure in the sense §3 defines. Both score low on isolation
 173 ($I = 0.18$ and 0.30, far below what the probe predicts), so isolation alone should not be used to
 174 choose a fix. A third case behaves as the information account predicts: at conv2, agent-relative
 175 concepts score low on *availability* too (0.338 against 0.808 at fc1, M3’s “winnable square” family).
 176 When a layer genuinely lacks a concept availability drops as well, and state threats do not. Everything
 177 below therefore concerns M3 / fc1, the one (agent, layer) where the dictionary has resolved anything

Table 3: Single-atom capture by concept shape, agent M3 / f c1, over all nine unsupervised dictionaries on that hook (3,209 concept verdicts). *pinned* = one location and one attribute pole. Within COUNT, holding agent-relativity fixed, changing only the logical form takes capture from 81.2% to 0.0%; holding form fixed and changing agent-relativity costs 2 points. solo = concentration; k_{90} and R^2 are medians.

basis	concept shape	agent-rel.	n	captured	solo	k_{90}	R^2
state	pinned	no	608	85.5%	0.974	1	0.763
count	pinned	no	532	81.2%	0.914	1	0.532
count [−]	pinned	no	532	62.2%	0.894	2	0.506
count	pinned	yes	532	79.0%	0.974	1	0.522
count [−]	pinned	yes	532	56.6%	0.862	2	0.419
count	OR over 4 attributes	no	133	0.0%	0.311	4	0.741
count[−]	OR over 4 attributes	no	133	0.0%	0.320	4	0.621
agent-rel.	OR over 8 poles	yes	171	0.0%	0.186	11	0.526

178 into atoms at all and so the only place where “what does it do with the concepts it still misses” is well
179 posed.

180 4.4 The residual wall is a disjunction wall

181 Our own earlier reading of these runs blamed *agent-relativity*: the concepts M3 / f c1 still misses
182 are the ones phrased from the mover’s point of view. A basis fixes logical form, agent-relativity and
183 prevalence at once. The comparison that separates them is *within* a single basis, and COUNT supplies
184 it: it contains both logical shapes and both roles (Table 3).

185 **Disjunction costs everything.** Within COUNT, among the concepts that are *not* agent-relative, pinned
186 ones are captured 81.2% of the time and disjunctive ones **0.0%** (0 of 133); COUNT[−] repeats it
187 independently on the opposite attribute poles, 62.2% $\rightarrow$ 0.0%. Pooled over all four bases: 73.2%
188 (pinned, $n = 2,736$) against 0.0% (disjunctive, $n = 437$, zero exceptions).

189 **Agent-relativity costs almost nothing.** Within COUNT, pinned agent-relative concepts are captured
190 79.0% against 81.2% for pinned state concepts (2 points, on 532 concepts each). Those agent-relative
191 concepts read the offered piece and have a median base rate of 0.0030, and are still captured four
192 times in five. Pooled over bases the gap is 9 points. Whatever the wall is, it is not agent-relativity.

193 **Nor is it prevalence.** A matched comparison settles it: restricting both arms to a common base-rate
194 window $[0.015, 0.050]$ leaves the effect intact, 71.4% captured for pinned ($n = 868$) against **0.0%**
195 for disjunctive ($n = 437$). The direction runs the wrong way at the level of shape too: the disjunctive
196 concepts are the *more common* (base rates 0.0316/0.0226) and the least captured, while the rarer
197 pinned concepts (0.0152/0.0029) are captured 68–77%. Critically, **the disjunctive concepts are not**
198 **information-poor** (Figure 2, left): they have the *highest* retention in the table: median $R^2 = 0.741$
199 for COUNT’s 4-way disjunctions against 0.532 for its pinned state concepts. The dictionary knows
200 the answer and has no atom for it.

201 4.5 The mechanism, and what capacity does not fix

202 Because each ground-truth concept is a known Boolean function of the board, we know how many
203 disjuncts it has *before looking at any dictionary*: one for a pinned concept, four for an “any threat
204 here” concept, eight for an agent-relative “winnable” one. That makes k_{90} (normally too brittle to
205 read, because noise in the R^2 tail pushes the crossing right) usable as a *distribution* compared against
206 a prediction it was not fitted to (Figure 2, right). The dictionary reconstructs a 4-way disjunction
207 from exactly four latents: over its 266 such concepts the mode of k_{90} is **4** (57.5%) and **not one is**
208 **below 4**, against a mode of 1 (60.6% of 2,736) for pinned concepts. This is a direct observation of
209 tiling rather than an inference from a threshold: the prediction is fixed by the concept definitions,
210 and the measurement was free to land anywhere in $[1, 64]$. The 8-way case is predicted less sharply
211 and behaves less sharply (mode 9, median 11, 5.3% below prediction, 21.6% at 20 or beyond), the
212 expected outcome for a concept that ORs over a larger and rarer set.

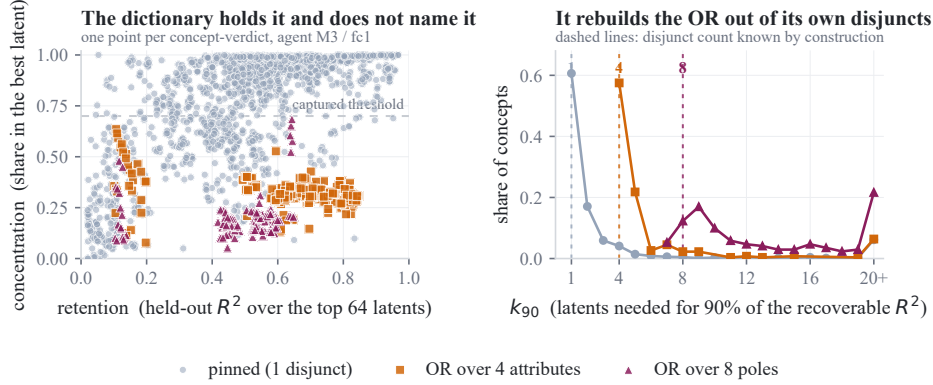


Figure 2: Agent M3 / fc1, one point per concept verdict over nine unsupervised dictionaries. **Left:** the same geometry as Figure 1 one level down: the disjunctive concepts sit far to the right (the dictionary holds the signal) and low (no atom carries it), while the pinned cloud sits above the captured threshold. **Right:** k_{90} against the disjunct count each concept has *by construction* (dashed). The modal cost of a 4-way OR is exactly four latents and nothing is cheaper; the 8-way case peaks at 9 with a long tail (21.6% at 20 or beyond, shown clipped).

213 The precondition bounds the claim: the signature appears *only* on M3 / fc1, the one cell where
 214 pinned concepts have modal $k_{90} = 1$. Everywhere else pinned concepts already need 6–30 latents,
 215 so there are no atoms for a disjunction to be tiled out of and k_{90} is large and unstructured. Tiling
 216 is conditional on atoms existing, and that precondition is exactly what conv2 and the two weaker
 217 agents lack (§4.3).

218 **Capacity does not fix it.** If the missing OR were a budget problem, a larger dictionary would buy it.
 219 Holding recipe, seed, hook and agent fixed and varying only d_{dict} across 4,096 $\rightarrow$ 16,384 $\rightarrow$ 32,768,
 220 the same 22 of 23 agent-relative threat concepts come out *spread* at every width, with median
 221 concentration flat at 0.19–0.20.

222 4.6 Controls, robustness, and one cell we do not quote

223 **The captured arm fires, and selection does not manufacture retention.** A three-way verdict is
 224 worthless if one arm never triggers. Across the panel captured fires 2,265 times in 12,151 verdicts
 225 (18.6%; spread 74.6%, absent 6.8%), including the state basis’s threat categories at 85.5% on M3 /
 226 fc1; the supervised positive control (slots anchored to the agent-relative concepts by an auxiliary loss)
 227 comes out 16 of 23 captured at concentration 0.76, against 0/23 for unsupervised dictionaries on
 228 identical rows. All 114 cells carry 100% random-control coverage and none is provisional. Separately,
 229 ranking latents on rows that overlap the rows R^2 is measured on is double-dipping the permutation
 230 null cannot see, since it shuffles labels *after* selection; a selection-aware null (permute first, then refit
 231 the whole pipeline) returns -0.0006 to -0.0013 where the real values are 0.058–0.656, so selecting
 232 64 of the 199–643 live latents buys no held-out R^2 from noise.

233 **Repairing the weakest dictionary.** One panel member (M3’s BatchTopK dictionary at conv2)
 234 was degenerate (99.0% dead latents, FVU 0.110 against 0.043/0.051 for the same recipe on the
 235 other agents), because the dead-latent auxiliary loss of Gao et al. [2024] was applied to TopK but
 236 not to BatchTopK in our code. A broken conv2 dictionary flatters our thesis, so we retrained it
 237 (FVU 0.110 $\rightarrow$ 0.0066) and repeated the analysis. Of 15 families **2 change sign, both per-cell,**
 238 **both toward the partition;** all twelve relational families are unchanged, per-cell isolation recovers
 239 $\sim 5 \times$ (0.108 $\rightarrow$ 0.527), and square-threat isolation moves only 0.115 $\rightarrow$ 0.178 against fc1’s 0.851.
 240 Represented by its *best* dictionary rather than its worst, conv2 is demonstrably competent at per-cell
 241 concepts and still fails on threats. All conv2 numbers above use the repaired dictionary except Table 1,
 242 whose pre-repair split counts understate the partition.

243 **Stability, and the one cell we do not quote.** Across a 4-condition $\times$ 3-seed grid (1,764 concepts),
 244 11.4% of concept verdicts flip between seeds, and 23 of 24 (agent, layer, basis) cells agree across
 245 three distinct recipes. The exception is M3 / conv2 / AGENT-REL., where retention sits at $1.6\times$ the
 246 untrained-network floor, close enough that the absent/spread label is a coin flip (0.39/0.65/0.39
 247 across seeds), so we quote no verdict for it. What all seven of its runs do agree on is the measurement:
 248 $1.6\times$ the floor at conv2 with concentration 0.10–0.34, against $14.6\times$ at fc1 on the same concepts
 249 and agent.

250 5 Related work

251 Layer specialisation in game-playing networks is established: Lovering et al. [2022] find in AlphaZero-
 252 for-Hex that short-term concepts probe best in final layers and long-term ones near mid-depth, and
 253 Du et al. [2025] report the SAE-based analogue in OthelloGPT. Both compare *probes* (or dictionaries)
 254 across layers; what we add is the comparison of *probe and dictionary on the same activations*, which
 255 separates “the layer knows it” from “the dictionary can name it.” Under a probe-only view §4.2 is
 256 invisible: the probe alone says both layers carry threats, the dictionary alone says only the bottleneck
 257 does.

258 Our disjunction result is, we think, feature absorption [Chanin et al., 2024] observed with exact
 259 ground truth. Absorption is diagnosed in language models as a general feature (“starts with S”)
 260 failing to fire because its children have taken the mass; here the general feature is a disjunction we
 261 can write down, its children are the disjuncts we can also write down, and we can count them; and its
 262 surviving an $8\times$ capacity increase reproduces their conclusion that size and sparsity are not the fix.
 263 The natural response is a dictionary whose units may stand in a parent/child relation (Matryoshka
 264 SAEs [Bussmann et al., 2025], hierarchical SAEs [Muchane et al., 2025]), and our measurement
 265 supplies an evaluation those architectures currently lack: a concept set with a *known* disjunct count,
 266 where success means a parent atom firing on the union of children that already exist and are already
 267 clean. This is a different brief from multi-dimensional or manifold-aware dictionaries [Engels et al.,
 268 2024, Park et al., 2024], the right response to a concept spread over an irreducible subspace, which
 269 also produces high availability with low isolation, but not the tiled-by-disjunct signature of §4.5.
 270 Finally, our per-family insistence echoes Karvonen et al. [2024]: board-game ground truth is valuable
 271 precisely because it is fine-grained, and averaging over the whole menu discards that.

272 6 Discussion and limitations

273 A low SAE score is not self-interpreting. On these agents low isolation means three different things,
 274 spanning a $3\times$ range ($I = 0.096\text{--}0.30$): agent-relative concepts barely present in the convolutional
 275 layer, state threats present in the layer but absent from its dictionary, and disjunctive concepts present
 276 in the bottleneck’s dictionary but split across their disjuncts. Availability separates the first from the
 277 other two, retention the second from the third, and only the third is a failure dictionary design can
 278 address. Reporting a probe baseline and a retention floor per concept family alongside any SAE
 279 evaluation is cheap and changes which fix you reach for.

280 It also narrows the brief. The strongest agent’s bottleneck already produces clean single atoms for
 281 every pinned threat ($k_{90} = 1$, median concentration 0.93); what is missing is not a better basis for
 282 those atoms but an *aggregating* unit above them. That points at the hierarchical arm of the literature
 283 rather than the manifold arm, with a sharp, pre-registrable test: a successful dictionary must move the
 284 4-way disjunctions from $k_{90} = 4$ to $k_{90} = 1$ without disturbing the pinned atoms they are built from.

285 **Limitations.** (i) **Decodability, not causality:** every number says a concept can be read out, not
 286 that the policy uses it; whether the clean pinned atoms are the ones the policy consumes is untested.
 287 (ii) **Retention is measured over 64 of 4,096 latents:** a concept recoverable only from hundreds of
 288 latents reads as low retention (the right reading for interpretability, not for information content). (iii)
 289 **The tiling signature is conditional** on the dictionary having resolved atoms at all, and holds on
 290 one (agent, layer) pair. (iv) **Linear availability only, and two layers rather than a depth sweep.**
 291 (v) **One game, one architecture family;** whether the partition and the disjunction wall transfer
 292 elsewhere is untested.

References

- Bart Bussmann, Patrick Leask, and Neel Nanda. Batchtopk sparse autoencoders. *arXiv preprint arXiv:2412.06410*, 2024.
- Bart Bussmann, Noa Nabeshima, Adam Karvonen, and Neel Nanda. Learning multi-level features with matryoshka sparse autoencoders. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025. arXiv:2503.17547.
- David Chanin, James Wilken-Smith, Tomáš Dulka, Hardik Bhatnagar, Satvik Golechha, and Joseph Bloom. A is for absorption: Studying feature splitting and absorption in sparse autoencoders. *arXiv preprint arXiv:2409.14507*, 2024.
- Davide Chicco and Giuseppe Jurman. The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation. *BMC Genomics*, 21(1):6, 2020.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- Jason Du, Kelly Hong, Alishba Imran, Erfan Jahanparast, Mehdi Khfifi, and Kaichun Qiao. How gpt learns layer by layer. *arXiv preprint arXiv:2501.07108*, 2025.
- Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not all language model features are one-dimensionally linear. *arXiv preprint arXiv:2405.14860*, 2024.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.
- Adam Karvonen, Benjamin Wright, Can Rager, Rico Angell, Jannik Brinkmann, Logan Smith, Claudio Mayrink Verdun, David Bau, and Samuel Marks. Measuring progress in dictionary learning for language model interpretability with board games. *arXiv preprint arXiv:2408.00113*, 2024.
- Kenneth Li, Aspen K. Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. *arXiv preprint arXiv:2210.13382*, 2022.
- Charles Lovering, Jessica Zosa Forde, George Konidaris, Ellie Pavlick, and Michael L. Littman. Evaluation beyond task performance: Analyzing concepts in alphazero in hex. *arXiv preprint arXiv:2211.14673*, 2022.
- Mark Muchane, Sean Richardson, Kiho Park, and Victor Veitch. Incorporating hierarchical semantics in sparse autoencoder architectures. *arXiv preprint arXiv:2506.01197*, 2025.
- Neel Nanda, Andrew Lee, and Martin Wattenberg. Emergent linear representations in world models of self-supervised sequence models. *arXiv preprint arXiv:2309.00941*, 2023.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. *arXiv preprint arXiv:2406.01506*, 2024.
- Senthoran Rajamanoharan, Tom Lieberum, Nicolas Sonnerat, Arthur Conmy, Vikrant Varma, János Kramár, and Neel Nanda. Jumping ahead: Improving reconstruction fidelity with jumprelu sparse autoencoders. *arXiv preprint arXiv:2407.14435*, 2024.
- W. J. Youden. Index for rating diagnostic tests. *Cancer*, 3(1):32–35, 1950.